

DAY 1

Overview of this course | Introduction to Artificial Intelligence & its applications

Introduction to Python | Exploring various python platforms | Installing & Working with Python IDLE

DAY 2

Introduction to Google Colab | Working With Colab – Some Basic Programs | Numpy Library | Matplot Library

Read & Write files | Image Resize | Grayscale Image | Gaussian

Blur | Threshold | Image Flipping

DAY 3

COMPUTER VISION

Introduction to Computer Vision & its applications | Overview on CV libraries & installation | Basic programs on Computer vision

PROJECT – Opening camera | Opening Camera Using IP | Eye Detection |

Smile Detection

DAY 4

PROJECT

Object Tracking based on Colour with OpenCV

Face Recognition with OpenCV

Recognizing Emotions from Faces

Gender and Age Prediction

DAY 5

DEEP LEARNING

Introduction to Deep Learning | Neurons & Activation functions | Deep learning libraries & how to install DL libraries

PROJECT

Real-time Object recognition using Pre-trained Model

Image Classification using Convolutional Neural Networks

DAY 6

PROJECT

Hand Gesture Recognition with Deep Learning

Detecting Plant Leaf Diseases using Deep Learning

Attendance System Face Recognition using DL & ML

DAY 7

PROJECT

Vehicle Detection and Counting

Driver Drowsiness & Yawning Detection

Licence Plate Recognition System

DAY 8

MACHINE LEARNING

Introduction to Machine Learning | ML Algorithms

PROJECT Medical Diagnosis System

NATURAL LANGUAGE PROCESSING

Natural Language Processing Essentials

PROJECT

Natural Language Processing-based & Chatbot Deployment on Telegram

DAY 9

SESSION :Sound & Audio Processing

PROJECT

Visualizing Sound , Spectrogram of Sound

Extracting text from image,video,handwritten notes

DAY 10

Machine Learning Algorithms: Build intuition for core supervised and unsupervised learning algorithms.

Linear Regression Algorithm – Gradient descent, cost function minimization, overfitting control

Logistic Regression Algorithm – Sigmoid activation, decision boundary formation, loss function

DAY 11

Support Vector Machine (SVM) – Margin maximization, kernel trick, role of hyperplanes

K-Means Clustering – Centroid update, convergence, distance measures

DAY 12

Deep Learning Algorithms: Understand how neural networks learn and generalize.

Artificial Neural Networks (ANNs) – Forward/backward propagation, activation functions

Optimization Algorithms – Gradient descent variants (SGD, Momentum, Adam), convergence behavior

Regularization & Normalization Techniques – Dropout, batch normalization, weight decay

Convolutional Neural Networks (CNNs) – Convolution operation, pooling, feature extraction

DAY 13

Text Preprocessing Algorithms – Tokenization, stemming, lemmatization, stopword removal

Word Representation Algorithms – One-hot encoding, Word2Vec (CBOW & Skip-Gram concepts)

Sequence Modeling Algorithms – RNNs, LSTMs, vanishing gradient problem

Attention Mechanism – Motivation for attention, concept of context vector and weighting

DAY 14

Computer Vision Algorithms + Transformers: Learn core algorithms for image understanding and transformer architecture fundamentals.

Feature Extraction in Vision – Edge detection (Canny), HOG, SIFT concepts

CNN Architecture Review – Understanding layers and receptive field

Transformer Architecture Overview – Self-attention, multi-head attention, positional encoding

DAY 15

Generative AI Algorithms: Understand algorithmic principles behind generative models and retrieval-augmented generation.

Generative Models Overview – Difference between discriminative and generative approaches

Variational Autoencoders (VAEs) – Encoder-decoder concept, latent representation

Diffusion Models (Intro) – Denoising process, iterative sampling, algorithmic intuition

Retrieval-Augmented Generation (RAG) – Retrieval mechanism, embedding store etc.